

AP CALCULUS AB - UNIT 1

Guided Notes 1.9: Connecting Multiple Representations of Limits



Name: _____ Class: _____ Date: _____

Learning Objectives and Essential Question

Essential question: How can we extract one-sided limits, two-sided limits, and point values from any representation?

- Extract one-sided limits, two-sided limits, and point values from any representation.
- Translate accurately among formula, graph, table, words, and notation.
- Distinguish exact conclusions from numerical estimates.
- Audit whether multiple representations are locally coherent.

Key Knowledge, Vocabulary, and Notation

Representation

A graph, table, formula, verbal statement, or symbolic statement that encodes mathematical behavior.

Local information

Limit questions depend on behavior arbitrarily close to the target.

Point information

$f(c)$ is a separate datum and may disagree with the limit.

Coherence

Different representations are coherent when they encode the same one-sided limits and point value.

Precision

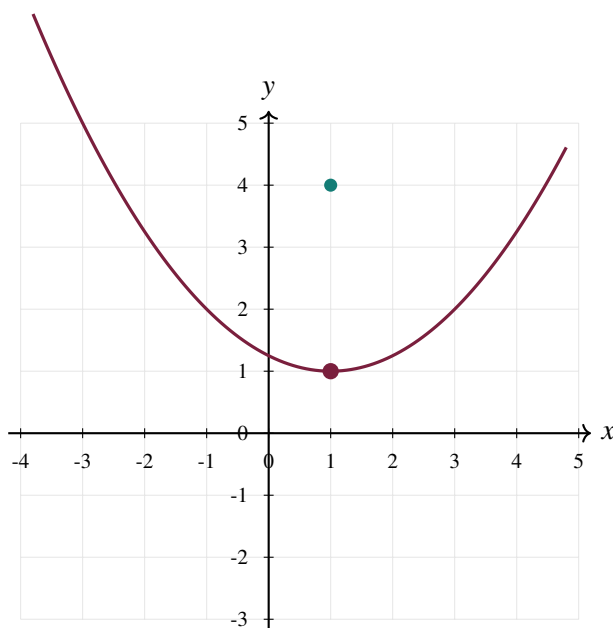
Algebra may establish an exact value; a graph or finite table generally provides an estimate.

Concept Development and Strategy

1. Create a fact ledger: left limit, right limit, two-sided conclusion, and point value.
2. When translating a formula into a graph, preserve excluded inputs and special assigned values.
3. For composite limits, include the direction from which the inner function reaches the outer target.

Shared Representation / Data

Representation Set R



x	-0.10	-0.01	-0.001	0.001	0.01	0.10
$p(x)$	2.31	2.0301	2.003001	1.997001	1.9701	1.71

The formula $p(x) = 2 - 3x + x^2$ for $x \neq 0$ and the assigned value $p(0) = 5$ describe a separate function with the same local pattern as the table.

Worked Example: Formula to graph facts

Let $f(x) = \frac{x^2-1}{x-1}$ for $x \neq 1$ and $f(1) = 6$. Describe the graph and state the limit.

Answer and Reasoning

For nearby $x \neq 1$, $f(x) = x + 1$, so the graph is the line $y = x + 1$ with an open circle at $(1, 2)$ and a filled point at $(1, 6)$. The limit is 2.

Worked Example: Table to notation

Use Representation Set R to write a limit statement at $x = 0$.

Answer and Reasoning

Values from both sides approach 2, so $\lim_{x \rightarrow 0} p(x) = 2$. The separate assigned value is $p(0) = 5$.

Worked Example: Directional composition

If $g(x) \rightarrow 1^-$ and a graph of F has $\lim_{t \rightarrow 1^-} F(t) = 4$ but $\lim_{t \rightarrow 1^+} F(t) = -2$, find $\lim F(g(x))$.

Answer and Reasoning

The inner values land at 1 from below, so the outer left-hand limit applies and the composite limit is 4.

Guided Practice and Discussion

Directions

Show the mathematical setup, preserve exact values unless a decimal is requested, and justify existence or interpretation statements with the appropriate definition or theorem.

1. A graph approaches 3 from both sides at $x = 2$ and has a filled point at 7. Write the limit and point value.

2. A table approaches -4 from both sides at $x = 1$. Translate this into notation.

3. Use Representation Set R to state the left-hand limit, right-hand limit, and two-sided limit at 0.

4. Use Representation Set R to state the assigned point value in the formula description.

5. A graph shows a hole at height 5, but a table rounded to the nearest integer shows 4 on the left and 6 on the right. Explain how both could describe the same limit.

6. Construct a formula whose graph has a hole at $(2, 5)$ and a filled point at $(2, -1)$.

7. Create five coherent representations for a function with $\lim_{x \rightarrow 1} f(x) = 4$ and $f(1) = -3$: notation, words, formula, graph description, and table description.

AP Reasoning Check

Multiple-Choice Check

Choose the best answer. Explain why at least one distractor is incorrect during class discussion.

1. Which pair of statements is coherent?

- (A) $\lim_{x \rightarrow 2} f(x) = 3$ and a graph approaching 5 from both sides (B) $f(2) = 7$ and a filled point at $(2, 7)$
(C) $\lim_{x \rightarrow 2^-} f(x) = 1$ and a left table approaching 4 (D) A formula equal to $x + 1$ near 2 and a hole at height 8

2. A table approaches 6 from both sides while the formula is undefined at the target. Which is the best conclusion?

- (A) The limit must be undefined. (B) The limit is estimated as 6 and the point value is undefined.
(C) The point value is 6. (D) The limit is infinite.

3. A graphing calculator displays a line through $x = 3$ for $f(x) = \frac{x^2 - 9}{x - 3}$. What information may be hidden?

- (A) A removable hole at $x = 3$ (B) A vertical asymptote at $x = 0$
(C) A jump at every integer (D) A horizontal asymptote at $y = 0$

Common Errors and AP Precision

- Keep the value of a function at a point separate from its limiting behavior near that point.
- State one-sided behavior explicitly whenever a two-sided limit or continuity claim depends on agreement from both sides.
- A complete AP justification names the relevant definition, condition, or theorem rather than reporting only a numerical result.

Exit Ticket

Can two functions have identical graphs at every displayed pixel and different limits at the target? Explain.

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